

IoT solutions for Electric Vehicles (EV) Charging Stations: A driving force towards EV mass adoption

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Abstract—The world is moving towards pollution-free traffic, and Electric Vehicles (EVs) are gaining more marketability worldwide. The global EV market will witness a very high growth rate in the coming future. As the EVs enter the market, together comes the basic necessity of charging stations to power those vehicles. Although there are a few charging stations, the count is much lower than the actual requirement. There are several challenges associated with traditional charging stations. The challenges are as mundane as theft, low speed of charging, the inability of power consumption monitoring, monitoring and maintenance, and under-optimization of charging time. However, an IoT-based charging infrastructure can streamline the performance of charging the EVs and come to rescue these challenges. IoT-aided charging could be much easier, faster, and more effective, taking us a step toward mass adoption of Electric Vehicles. The Article proposes IoT based application framework with cloud gateways and data transformed for optimisation of the dimensions of electric vehicle charging as a real time solution.

Index Terms—Electric-vehicles (EV), IoT- Internet of Things, EV charging stations, IoT Aided Charging of EV, EV market

I. INTRODUCTION

The electric vehicle industry is undergoing a technological shift with twofold objectives: to enhance EV adoption and provide and manage the charging infrastructure. EV policies already enacted support the installation of charging stations with underlying developing regulatory initiatives for managing an electric vehicle fleet. The electric vehicle is powered by an electric motor rather than an internal combustion engine which requires fossil fuels for power generation. Hence, the mass adoption of these EVs will be one major step towards addressing the issue of global warming, depletion of natural resources, air pollution, etc. Different EVs have different working mechanisms. However, they work similarly—the EVs function on an electric motor powered by batteries, which are occasionally charged. For the mass adoption of these vehicles, infrastructures are being set up at multiple locations worldwide as there are numerous advantages tagged along with it, like fuel economy, costs, energy

security, and technological advances. India will be the fourth largest market for EVs by the year 2040 [8]. The rapid growth of Electric Vehicles has led to the proliferation of EV charging stations, and the rate at which EVs can be charged is becoming one of the key differentiations in this emerging market.

A. IoT solutions for EV charging stations

The Internet of Things is a network of connected objects embedded with sensors, software, and other technologies for exchanging data and information. These intelligent devices collect massive amounts of data continuously stored in the Internet of Things cloud and studied for understanding human behavioral patterns of usage and leveraged for optimization. IoT is one of the most booming and significant technologies of the 21st century as its solutions have helped improve the control of operations in every sector by offering seamless communication and intelligence pooling.

B. IoT in the automotive sector

The emergence of IoT in the automobile industry was initially predicted as a futuristic theoretical concept. However, today we are already here working on the possibilities of connected cars, intelligent charging infrastructure, and dynamic load management, which outlines the fact that IoT is a disruptive technology with the potential to change our lives today. Nowadays, automobile buyers seek an intelligent vehicle to connect and interact with. The arrival of IoT in the automotive industry has opened new avenues for carmakers to build sustainable next-generation vehicles and charging solutions to expand the market horizons.

C. IoT solutions for EV charging stations

Several challenges affect smooth functioning when rolling out an EV charging infrastructure, such as station monitoring and management, accurate payment analysis, energy management, and theft management. With IoT sensors and software, EV charging stations can become smart, connected, and easily accessible for maintenance. IoT-powered applications can help locate the nearby charging stations; intelligent devices can monitor and manage the events on the platform; collect and analyze

data in real-time, which can help act security optimization and theft management. There are various other intelligent solutions that IoT can provide for the deployment and management of EV charging stations and amplify the EV market's growth, which will ultimately help the country move towards the electrical shift.

II. OVERVIEW OF ELECTRIC VEHICLES

An electric vehicle (EV) is an automobile that uses one or multiple electric motors for propulsion. Since 2010, there has been exponential growth in EV penetration. The number of electric cars globally has increased from 17,000 to over 10 million in 2021[1] Global Sales of Electric Vehicles are evaluated to Grow at 22 percent CAGR where the Market Share will rise to USD 700 Billion growth in this industry by 2028 [2].

A. Traditional vehicles and Electric Vehicles

The significant difference between conventional and electric vehicles is the Source of Power. The Energy source for Traditionally is Gasoline, Propulsion mechanism Engine, with CO₂ emissions with fuel facilities as petrol pumps and cruising distances are long with high fuel costs and advanced technologies developed over last 100 years. While electric vehicles have heat emission, shorter distance travel and low fuel cost as it runs on electric motors and yet the technologies are developing.

B. BEV (Battery Electric Vehicles)

These cars are charged only from an electric battery. Fully-electric cars emit no toxic CO₂ from their tailpipes. They are the cleanest vehicles on the track and are entirely fuelled by chargeable electric batteries. As a cost-effective option, businesses are rapidly transitioning to fully electric cars. They are dependable, simple to maintain, and enjoyable to drive, with various choices to fit any budget.

C. PHEVs (Plug-In Hybrid Electric Vehicles)

PHEVs have both an electric battery and a combustion engine. The battery is recharged from a charging plug and provides between 20 to 30 miles of pure electric transportation in the area and hundreds of miles of gasoline or diesel driving.

D. HEVs (Hybrid Electric Vehicles)

Hybrid Electric cars have both an electric battery and a combustion engine. The battery is charged using the combustion engine. HEVs blend the advantages of reduced tailpipe emissions and good fuel economy with the efficiency and range of traditional vehicles. BEVs are among the three stated types, which can be called fully electric vehicles. Other than motors and batteries, this electric vehicle type has no other means of propulsion. These cars produce no emissions. An external source of electricity, such as a DC quick charger or an AC charger, is used to charge the battery. Using an AC charger takes

an average of 8 hours to charge a BEV while using a DC quick charger completely; this time, it can be cut in half.

III. TYPES OF BATTERIES IN EVS

There are mainly three varieties of rechargeable batteries used in EVs:

- 1) Lead-acid batteries: Lead-acid batteries are the oldest, built in 1859 and still used today. They are made of re-cyclable materials and are a kind of wet cell battery that holds a mild sulfuric acid solution. Lead-acid batteries have the advantage of being less expensive and have been in operation for many years. This battery form has a three-year life cycle and needs electrolyte level inspection. They supply 25-50 percent of the vehicle's total mass with ample electricity.
- 2) Lithium-ion batteries: The most commonly used battery in electric cars is the lithium-ion battery, which is prohibitively costly. According to some researchers, this is the nearest research to developing an optimal rechargeable battery. The weight of a Li-Ion battery is one-third that of a standard battery. With an energy rate of 140+ WH/kg, Li-Ion batteries have an advantage over others. Every lithium-ion cell produces around 3.6 V of voltage. Most consumer lithium-ion batteries charge 4.2 volts per battery, with a resistance of around 50 millivolts per cell.
- 3) Nickel metal hydride: These batteries were introduced to the consumer market in the late 1980s. They have a 68 WH/kg value and a 60 to 120 WH/kg size with a potential voltage of 1.2V, indicating the amount of energy in the cell. These batteries have a good track record in electric vehicles. With these batteries, one would expect their vehicle to go above 100,000 miles and have a battery life of 5-7 years. However, there are a few disadvantages associated with them: Inefficient charging, self-discharge.

IV. EV MARKETS

Electric Vehicles (EV) are one option to reduce air pollution and greenhouse gas emissions caused by Internal Combustion engines. In the past decade, the worldwide EV industry has advanced dramatically. Globally, the number of light electric vehicles reached 2.2 million units by the end of 2019. And the biggest markets have slowed down on account of Covid where the mobility took a break with several countries observing a lock down situation [3],[4].

While Tesla is the world leader in electric vehicles, Ola Electric is seeking to become a significant participant in the Indian EV market. Ola Electric may not be considered a pioneer of the Indian electric vehicle revolution and parallels global EV leader Tesla. If confirmed at launch, it

might be one of the country's most cost-effective electric scooter possibilities with scalability as the bigger challenges [7]

V. BARRIERS TO MASS ADOPTION OF EVS

A. Battery Charging time

Although the charging time for different EVs is variable, filling up a gas tank might take up to three minutes at the gas station, while recharging an electric car's battery takes much longer. Sometimes take up to 20 hours to fully charge.

B. Government incentives

The retail price, usually higher than its internal combustion engine equivalent even after rebates, is a significant obstacle to affording these EVs. Many people still see electric vehicles as a status symbol. Although many cost-saving mechanisms associated with car ownership support an argument for cost parity and cost savings over the lifespan of vehicle ownership, many customers find the initial capital cost impossible to resolve.

C. Lack of Charging Infrastructure and Range Anxiety

As the number of electric vehicles on the road grows, so does the need for charging infrastructure. A lack of charging facilities may make electric vehicles less convenient and contribute to range anxiety, resulting in fewer people adopting the technology. Concerns were expressed in a survey conducted by the Union of Concerned Scientists, which found that range anxiety was the top issue when considering the mass adoption of Electric Vehicles. Because of this concern, the BEV target market is constrained. In March 2020 [8]. However, due to a lack of EV charging infrastructure, electric bikes and electric cars generally have been hard to come by for technological advancements at global levels.

D. Insufficient Standardization of EV charging structure

The necessity for standardization of electric car charging stations has been highlighted by points such as the growth of the EV industry and variations in charging loads. Some EV charging stations may only function with a specific voltage. AC charging stations, for example, give a voltage of 120V AC at level 1 charging stations and 208/240V AC at level 2 charging stations. DC charging stations, contrarily, use 480V AC to provide rapid charging. Particular countries have different quick charging standards. Because India has yet to achieve standardization in fast charging systems, the Indian government has ordered the installation of both CHAdeMO and CCS methods. However, the government in May 2021 said that the release of the standards would formally take place in the coming two months once the field and durability tests of sample products are completed [6].

VI. OVERVIEW OF ELECTRIC VEHICLES (EV)CHARGING STATION (CS)

A. EV Charging Market

An electric vehicle charging station is a piece of infrastructure that attaches an electric vehicle (EV) to a power supply so that it can be recharged. Electric power providers or commercial companies have charging stations in public parking lots, or retail shopping malls, also called electric vehicle supply equipment (EVSE). These stations have unique connectors that comply with several different electric charging requirements. The cost of using an EVSE ranges from monthly or annual fares to per-kWh or hourly fares. However, charging stations are mainly free to use and funded by municipal governments. EVSEs come in a variety of sizes and charge at various speeds [9]. The global EV Charging Station market is worth 39.7 billion dollars and is led by key players, including Charge Point (US), Shell (Netherlands), Blink Charging (US), BYD (China), and Tesla (USA) (US)[10]. With the outbreak of COVID-19, the automotive industry faced challenges like a lack of raw materials and labor [5]

B. EV Charging Market- Recent Developments

Magenta, based in Mumbai, has taken charge of the country's largest public EV charging station, with up to 21 charging stations, launched in July 2021 [11]. Tesla purchased German ATW automation, a firm specializing in battery components, in September 2020 [12]. In July 2020, ABB began working on a USD 30 million facility in the Netherlands to manufacture electric vehicle chargers for global demand [11]. To boost its E-mobility portfolio, ABB bought Charge Dot, a Chinese EV charging service, in March 2020 [11]. In September 2020, ChargePoint released the Home Flex charger (50amp). Tesla released the 3rd generation supercharger in March 2019 [9].

C. Types of EV chargers

There are three main varieties of EV charging— rapid, fast, and slow. These are the available power outputs, and thus charging speeds, for an EV with a kilowatt rating (KW). Each charger has its unique set of connectors suitable for low- or high-power applications for charging via AC or DC.

D. Rapid Chargers

Rapid chargers are the quickest way to charge an electric vehicle, and they are commonly seen at highway rest stops or along major highways. EVs can be charged up to 80 percent capacity in as little as 20 minutes, depending on the model, but on a typical 50 kW rapid charge stage, an average EV will take about an hour. Rapid AC chargers use the Type 2 charging standard and have 43 kW (three-phase, 63A) power. Depending on the model's battery power and starting state of charge, rapid AC units will charge an EV in minutes.

E. Fast Chargers

Usually, fast chargers are rated at 7 kW or 22 kW. (Single-or three-phase 32A). While most quick chargers

use AC power, some networks install 25kW DC chargers with CCS (Combined Charging System) or CHAdeMO (CHArge de MOve) connectors. Charging times differ depending on device speed and vehicle, but a 7kW charger can recharge a compatible EV with a 40kWh battery in 4-6 hours and a 22kW charger in 1-2 hours. If a cable is tethered to the unit, it can only be used by models with the connector style. Untethered units are thus more adaptable and, with the proper cable, can be used by any electric vehicle. The charging rates when using a fast charger will vary on the car's onboard charger, as not all models can accept 7 kW or more.

F. Slow Chargers

The majority of slow charging units are rated at up to 3 kW. In practice, slow charging occurs at speeds ranging from 2.3 to 6 kW, with 3.6 kW slow chargers being the most common (16A). The car can usually draw 2.3 kW (10A) while charging on a three-pin socket, whereas most lamppost chargers are capped at 5.5 kW due to current facilities. Charging times differ depending on the charging device and the EV being charged, but a complete charge on a 3 KW unit takes 6-12 hours on average.

VII. CHARGING TIME OF BATTERIES VIA CHARGERS

The charging time is primarily influenced by the battery's capacity, power density, and charging power. The more power a battery has, the more charge it can store. The battery can tolerate more charge/unit time with a higher power density for the size of the tank opening. Higher charging power provides more energy per unit of time. When the battery is low on power, the charging mechanism is most effective; charge acceptance slows as the battery approaches saturation, like when we are hungry, we gobble up food fast. Ultra-fast charging, also known as boost charging, must be carried out under specific circumstances. The temperature of a battery also influences how quickly it can be powered. Li-ion batteries often prefer a cold storage setting, but the charging and discharging work better at a higher room temperature [13].

VIII. SOURCES OF POWER FOR CHARGING STATIONS

There are mainly two types of charging using electricity, AC (Alternating Current) and DC (Direct Current). DC charging is commonly utilized in the street, offices, and service stations, while AC charging is used at home. Alternative sources of power for EV charging stations:

A. Motion Charging

Several companies have begun to develop devices that charge batteries using human motion. Tremont Electric's ex-ample uses a magnet between two springs to charge a battery while the unit is pushed up and down, such as while walking. Such products have not yet proven to be commercially successful.

B. Induction-powered charging-

Electromagnetic induction is used to charge batteries in inductive battery chargers. A charging station transmits electromagnetic energy to an electrical device through inductive coupling, which stores the energy in the batteries. There are no metal connections between the charger and the battery, so this is possible.

C. Pulse charging

Some chargers use pulse technology, which feeds the battery a voltage or current pulse sequence. The rise time, pulse duration, pulse repetition rate as the frequency, and amplitude of DC pulses are all precisely regulated. This technology is said to work with any type of battery, including automotive and valve-regulated batteries, regardless of size, voltage, power, or chemistry.

D. Solar charging

Solar chargers can convert light into low-voltage DC. They are usually lightweight but can be mounted permanently as well. Solar panels are also known as fixed mount solar chargers. Even though portable solar chargers rely solely on the sun for power, they can, may varies with technology, be used in low-light situations. While some solar chargers, depending on the wattage, can fully recharge batteries, portable solar chargers are frequently utilized for trickle charging.

IX. SWOT ANALYSIS OF BEV CHARGINGINFRASTRUCTURE

A. Strength

Achieve synergies with existing infrastructure. Scalable infrastructure investment opportunity exists. Home charging is availability. The flexibility of renewable electricity integration. The efficiency of the electric supply chain and Low charging costs.

B. Weakness

Suitable for passenger cars and LDV only. No inherent seasonal electricity storage. Long charging time and varied charging plug types. High Initial cost for Fast chargers

C. Opportunity

The new business model for enhanced charging operations Advanced worldwide infrastructure setup would lead to more significant visibility opportunities. Electricity as an established energy carrier Achieve synergies with renewable electricity development Multiple EV owners can be served at the same time with charging. Allow suppliers to monitor or charge for the electricity that is consumed easily. Falling lithium-ion battery prices When the EV is not charging, the facility may have additional current or connection monitoring measures to disconnect the electricity.

D. Threat

Uncertain consumer behavior. A vision of changing the felling process is blurry. Uncertain impacts on distribution grids

X. IOT SOLUTIONS FOR SMAREVCHARGING STATIONS- AN OVERVIEW

When an automobile is ‘smart charging,’ the charger is basically ‘communicating’ with your car, the charging operator, and the energy company via communications networks. To put it another way, anytime the electric vehicle is plugged in, the charger provides critical data to the company so they can optimize charging. As a result, smart charging allows the charging operator to control how much energy is sent to any plugged-in electric vehicle. Smart charging also keeps charging operators from going above their building’s maximum energy capacity, determined by local grid capacities and the energy tariff they choose. This saves time and money for everyone, and most importantly, it saves energy, allowing us to protect the planet’s limited resources better. [14] However, how do we facilitate this integration? That is where comes the Internet of Things. The Internet of Things is a network of connected objects embedded with sensors, software, and other technologies for exchanging data and information. There are numerous benefits of deploying charging stations embedded with IoT sensors, and the success of which could lead to a huge jump towards the mass adoption of Electric Vehicles.

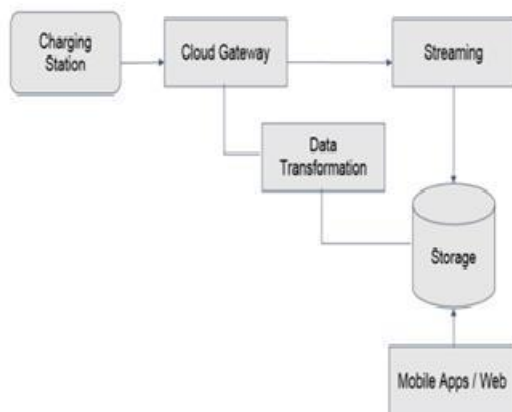


Fig 1. IoT-driven Charging Infrastructure.

XI. IOT SOLUTIONS FOR EV CHARGING STATIONS.

A. IoT Sensors

IoT sensors can be found in our homes, companies, shop-ping malls, hospitals, and everywhere. They are embedded in cell phones and play an essential part in the Internet of Things (IoT). Sensors have been around for a long time. However, the Internet of Things (IoT) and its industrial cousin, the IIoT (Industrial Internet of Things), are pushing sensor usage to new heights. There are many

IoT sensors and even a more significant number of applications and use cases.[15] • Optical Sensors • Motion Sensors • Proximity Sensors • My things IoT sensors • Infrared Sensors

B. Smart Meters

A smart meter is an electronic device that tracks and records information such as electric energy consumption, voltage levels, current, and power factor. These meters send data to consumers to help them better understand their usage behavior and power suppliers for system monitoring and customer requests. Intelligent meters often measure energy in real time and send out reports regularly throughout the day. The utility’s information technology, which combines intelligent meter networks with utility applications like billing and CIS, is another critical technology for intelligent meter systems. Due to these meters’ limited computational assets and long operational life, implementing security mechanisms to safeguard these devices from malicious attacks has proven difficult. [16]

C. IoT Analytics

IoT stream analytics can be used to turn data into a database. Businesses rely on three sorts of IoT analytics from their chosen IoT platforms.

- Descriptive Analytics- Descriptive analytics is the most fundamental type of analytic insight, allowing users to describe and combine IoT data.
- Predictive Analysis- By evaluating past data, predictive analytics predicts future data and actions.
- Prescriptive Analysis- Prescriptive analytics aids businesses in determining the best course of action for the future. Prescriptive analytics employs techniques such as image processing, machine learning, and natural language processing.

Based on vehicle and charging station telemetry details, the current battery state and power supplied by the charging station can be known. Users can have their battery status and time to the maximum charge shown in mobile and online applications. EV drivers can utilize maps offered by mobile apps/web apps to locate and reserve the nearest charging station. [17]

D. IoT Applications

An application would be integrated into EVs to build an intelligent energy management network. The application will connect EV batteries to grid-connected smart chargers, allowing them to act as a distributed micro-grid. The intelligent framework could safely and effectively track and manage energy. The intelligent framework could safely and effectively track and manage energy and find and book the closest CS via apps. These services combine public charging, home charging, energy, and new intuitive

features to make EV charging a simple part of our daily routine. EV charging user interfaces are effortlessly integrating mobility-as-a-service elements, and public and business car share options.[18]

E. Cloud Storage

In order to monitor charging points remotely while scaling up EVC infrastructure and managing large networks, these networks must be 'connected.' This solution seeks to provide numerous features, such as monitoring and to evaluate charger usage, setting prices, gaining remote access to chargers, and resolving incidents, all of which will significantly reduce operational expenses and improve operational efficiency. An EV cloud platform could enable this facility to enhance economic, social, and environmental value. In this implementation, the charging station sends a request to the CIO's server for reference input, giving the CIO server the user's credentials and identification and charging parameters in this initial request. The proposed method helps decarbonize road transport by combining EV charging and cloud-based IT solutions as an integrated system. The CIO's charging infrastructure management service can now communicate with any of this user's other management services. The CIO server responds to the request with a charging plan based on a P(t)-curve, which is then sent to the EV. Furthermore, the solution promotes EV market adoption, advancing electric transportation's social and environmental benefits.[19]

XII. EFFECTS AND OUTCOMES FOR IOT SOLUTIONS

A. Security Analysis

The Internet of Things (IoT) now plays a significant role in several fields through automating applications. Various wireless IoT sensors exist to detect power theft, which is on the rise in both home and industrial settings, posing a threat to the charging infrastructure. PIR sensors, for example, are electronic sensors that measure motion by measuring the quantity of infrared radiation emitted by objects. This hypothesis is used to detect an intruder who is stealing or misusing the charging components. Remote theft monitoring could thus be conceivable and feasible by installing or embedding those sensors in the chargers. [20]

B. Battery Monitoring (State of Charge Estimation)

The issue of range anxiety is addressed by tracking the battery's State of Charge using a phone application and finding and booking nearby charging spots.

C. Energy Management

The IoT sensors continuously measure the supply power and detect when the EV is fully powered. The EV charging process is halted, the data is sent to the IoT network (to keep track of the transaction), and power is passed on to the following EV in line. Based on the drivers' consumption profiles and statistical data about

their behavior, a priority/utility-based resource scheduler can be utilized to strengthen the benefits/utility of the power delivered, recorded from previously stored data, average power needed, the average count of hours before the EV is unplugged.[18]

D. Remote Monitoring and Maintenance

Alleviating the grid's growing stress requires optimal charging timing for electric vehicles. If not done ahead of time, it will result in a voltage and frequency imbalance, leading to grid collapse and blackout. The platform centralizes all user/system authentication, authorization, and auditing processes. It also gathers the user's usage behavior to predict the current charging process's power consumption. [20]

E. Autonomous Payments

EVs encounter several challenges when traveling abroad due to the time and effort required to obtain charging cards from foreign operators. With an identity management mechanism and a safe environment laid out via the Internet of Things, apps can be used for mobile ticketing systems, paying motorway tolls, and automobile device services. For example, charging stations in the United Kingdom will be outfitted with NFC payment technology. The Management Unit gets data from the network and instructs the power management unit to begin the process of charging, which can take some time because of the optimized power distribution between the chargers. [14]

XIII. DISCUSSIONS

A scarcity of charging stations may make electric vehicles inconvenient to adopt at mass. This paper suggests solutions for Electric Vehicle Charging stations leveraging the Internet of Things. To address the issue of Range anxiety and EV mass adoption on a more significant run, the Internet of Things (IoT) is a promising technology. The development of EV charging stations is necessary for the mass adoption of these electric vehicles, and IoT has played a significant role in the rapid development of those. The end-to-end charging station powered by IoT has a bright possibility due to enhanced electrical characteristics. Although there is a risk of damage to the battery pack in case of overcharging or deep draining circumstances, and uncontrolled charging can adversely disrupt the operation, the continuous monitoring solution will provide an edge over here. The chargers must be updated automatically by utilizing IoT and charging the automobiles. Moreover, intelligent charging stations also determine arriving EVs to admit and schedule them according to their charge capacity and

provide optimal options that ensure a seamless experience and lead to mass adoption.

REFERENCES

- [1] IEA Report 2020, "Global EV outlook 2020," 2020.
- [2] Facts and Factors Report 2021, "Global Sales forecast of Electric Vehicles," 2021.
- [3] McKinsey Report on "EV market," 2020.
- [4] G. Marine and L. Paoli, "EV market - covid effect," 2021.
- [5] EV global outlook, I. E. Agency, "2021".
- [6] PTI Report, "Indian Standards for charging point for EVs," 2021.
- [7] EV Leaders- H. Auto Report 2021, 2021.
- [8] Bloomberg, "Range Anxiety caused by lack of charging stations avail-ability," Press Trust India, 2020.
- [9] F. B. Insights Magazine Article, "Ev charge mkt," 2019.
- [10] Report Linker, "EV charging station key players of industry," 2021.
- [11] PTI Report 2021, "EV Charger Statistics," 2021.
- [12] Reuters Report 2020, Statistics, 2020.
- [13] Y. Miao, P. Hynan, A. Von Jouanne, and A. Yokochi, "Current lion battery technologies in electric vehicles and opportunities for advance-ments," MDPI, vol. 12, no. 6, pp. 1–20, 2019.
- [14] P. Arunkumar and K. Vijith, "IOT enabled smart charging stations for electric vehicle," Int. J. Pure Appl. Math., vol. 119, no. 7, pp. 247–252, 2018.
- [15] A. Karpenko et al., "Data exchange interoperability in iot ecosystem for smart parking and EV charging," Sensors (Switzerland), vol. 18, no. 12, 2018.
- [16] HM Government Department for Transport, "Electric Vehicle Smart Charging meters," no. July, pp. 1–67, 2019.
- [17] M. Adhikari, L. P. Ghimire, Y. Kim, P. Aryal, and S. B. Khadka, "Identification and Analysis of Barriers against Electric Vehicle use," Sustain., vol. 12, no. 12, pp. 1–20, 2020.
- [18] J. L. A. Jose P. Martins, Joao C. Ferreira, Vitor Monteiro, Jose A. Afonso, "IoT and Blockchain Paradigms for EV Charging System," MDPI, no. 9710, pp. 147–182, 2019.
- [19] M. Mierau, S. Fey, R. Kohrs, and C. Wittwer, "Communication solutions for a cloud-based charging management system for a fleet of shared-use electric vehicles," World Electr. Veh. J., vol. 6, no. 4, pp. 1125–1135, 2013.
- [20] R. Ravi, U. Surendra, N. Shreya, and R. Ravi, "EasyChair Preprint Comparative Analysis of Various Techniques of IOT in Electric Vehicle" 2020.